
Definition: Second-degree Taylor Polynomial of a function of two variables, $f(x, y)$

For a function of two variables $f(x, y)$ whose first and second partials exist at the point (a, b) , the **2nd-degree Taylor polynomial** of f for (x, y) near the point (a, b) is:

$$f(x, y) \approx Q(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) + \frac{f_{xx}(a, b)}{2}(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{f_{yy}(a, b)}{2}(y - b)^2 \tag{5}$$

If we have already determined $L(x, y)$, we can simplify this formula as:

$$f(x, y) \approx Q(x, y) = L(x, y) + \frac{f_{xx}(a, b)}{2}(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{f_{yy}(a, b)}{2}(y - b)^2 \tag{6}$$

b. $f(x, y) = xe^y + 1$ for (x, y) near the point $(1, 0)$

5) $f(x, y) = x^2y + y^2, \quad P(1, 3)$